

Deep Learning: Course Plan

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Some Good Resources for DL

- Duda and Hart, "Pattern Classification" Wiley, 2001.
- C. M. Bishop, "Neural Networks for Pattern Recognition", 1995.
- B. Yegnanarayana, "Artificial Neural Networks", PHI, 1999.
- C. M. Bishop, "PRML", Springer, 2006.
- Michael Nielsen "Neural Networks and Deep Learning" Open Book.
- Ian Goodfellow, et. al. , "Deep Learning", MIT Press, 2016.
- Charu Agarwal, "NNDL", Springer 2018.
- Mithesh M. Khapra, "Deep Learning Part I and II" NPTEL Course.
- Ali Ghodsi, "Deep Learning", University of Waterloo Course.
- C. M. Bishop & H. Bishop, "Deep Learning", Springer, 2024



Theory Course Information

- Course Title : Deep learning (DL)
- Course Code : CS 451
- Venue : C303
- Tue : 9-10.30 am
- Wed : 9-10.30 am (Lab / Course Project / Extra Slot)
- Thu : 9-10.30 am
- Online slot : Thursday (7-8.30 pm) (Extra Slot)



Theory Course Evaluation (3-0-0-4-4)

- Course Instructor: S R M Prasanna
- First instructional day (T_0)
- Quiz-I (5%) : $T_0 + 4W$; Wednesday, 28th August 24, 9-10 am
- Midsem (20%) : As per calendar, $T_0 + 7W$;
- Quiz-II (5%): $T_0 + 11W$, Wednesday, 6th Nov 24, 9-10 am
- Endsem (30%): As per calendar, $T_0 + 14W$;
- Lab implementation (15%) (CE)
- Course Project (15%) (CE)
- Endsem viva-voce (10%)
- Attendance Policy : As per institute norms



Syllabus

- **Introduction:** Artificial intelligence, pattern recognition, machine learning, & deep learning; neural networks vs deep learning; applications of deep learning
- **Review of Neural networks:** Biological neuron & neural network; artificial neuron, neuron models, artificial neural networks, learning & learning laws, feedforward, feedback and hybrid artificial neural networks; hard problems, hard-learning, back-propagation learning algorithm
- **Deep feedforward neural networks (DNNs):** Training deep network - greedy layerwise training; training belief network, deep feedforward neural network; significance of training DNNs in handwriting recognition and speech recognition tasks



- **Convolutional neural networks:** CNN vs other neural networks; basic modules of CNN; training and testing of CNN; CNN as feature extractor; CNN as classifier; significance of CNN in object classification task
- **Sequence models:** Sequence models vs discriminative / generate models; Recurrent neural network (RNN); Long short term memory (LSTM) network; gated RNN (GRU); Seq2seq neural network; Attention Modeling; Transformers; applications of sequence models
- **Generative models:** Generative vs discriminative models, variational autoencoder (VAE), generative adversarial networks (GAN), and generative pre-trained transformer (GPT); applications of generative models.



Lab Implementation

- Individual
- Some 10 implementations of different neural network and deep learning models
- Demonstrating the understanding of working principle
- Artificial neurons & learning
- Expt to illustrate differences among learning laws
- Feedforward network using perceptron learning
- Feedforward network using back-propagation learning
- Belief network training
- DNN, CNN, RNN, GRU, LSTM, VAE, GAN, Transformer

Course Project

- Team of 2 members
- Fair policy of declaration of contribution
- Speech processing research based projects
- Problem titles will be floated
- Discuss and choose one
- Working with my past and present research scholars, and some collaborators



Quiz, Midsem, Endsem

- Written exam (1, 2, 3 hr)
- Descriptive, derivation, problem solving
- Not direct questions
- Mind twisters

